

Warnaar's Conjecture 2 at $k = 3$ for the balanced profile $(3, 3, 2)$: a complete proof via Warnaar's finite form, a Kanade–Russell contiguous relation, and Uncu's modulo-11 theorem

Seed 2, Layer 3, Round 2 (loop experiment 2026-07-04)

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Abstract

We prove the balanced case $c = (3, 3, 2)$ of Warnaar's Conjecture 2 (the A_2 Andrews–Gordon fermionic-form conjecture for cylindric partitions) at level $d = 8$ (modulus 11), by chaining three proved results: Warnaar's finite-form Proposition ($a = -1$, $k = 3$), a single instance of the Kanade–Russell contiguous relation R_3 , and Uncu's 2024 theorem expressing $H_{(3,3,2)}$ in the series $S_{11}(\rho|\sigma)$. As a corollary the coefficients $Q_{n,(3,3,2)}(q)$ are manifestly nonnegative: the first proved member of the positivity core at $d = 8$, and the first proved case of Conjecture 2 at $k = 3$.

1 Setup and statement

Throughout, $(q; q)_n = \prod_{i=1}^n (1 - q^i)$, and $\begin{bmatrix} n \\ m \end{bmatrix}$ is the q -binomial coefficient (zero unless $0 \leq m \leq n$). For a profile $c = (c_1, c_2, c_3)$ with $c_1 + c_2 + c_3 = d$, let $F_c(z, q)$ be the Corteel–Welsh cylindric-partition generating function, and set

$$G_c(z, q) = (zq; q)_\infty F_c(z, q), \quad H_c(z, q) = \frac{(zq; q)_\infty}{(q; q)_\infty} F_c(z, q) = \frac{G_c(z, q)}{(q; q)_\infty},$$

$$Q_{n,c}(q) = (q; q)_n [z^n] G_c(z, q).$$

Warnaar's conjecture asserts $Q_{n,c}(q) \geq 0$ coefficientwise for all n, c . For the balanced profile $c = (k, k, k - 1)$, $d = 3k - 1$, Warnaar's Conjecture 2 predicts the explicit fermionic form (no linear terms): with $m_k := 2n_k$,

$$G_{(k,k,k-1)}(z, q) \stackrel{?}{=} \text{FERM}_k(z, q) := \sum_{\substack{n_1, \dots, n_k \geq 0 \\ m_1, \dots, m_{k-1} \geq 0}} \frac{z^{n_1} q^{n_k^2}}{(q; q)_{n_1}} \prod_{i=1}^{k-1} q^{n_i^2 - n_i m_i + m_i^2} \begin{bmatrix} n_i \\ n_{i+1} \end{bmatrix} \begin{bmatrix} n_i - n_{i+1} + m_{i+1} \\ m_i \end{bmatrix}. \quad (1)$$

This is proved for $k = 1, 2$ (Warnaar); $k = 3$ ($d = 8$, modulus 11) was open.

For $m = 11$ ($k = 4$ in Uncu's indexing), $\rho, \sigma \in \mathbb{Z}^3$, define (Uncu 2024, eq. (Sm1))

$$S_{11}(z; \rho|\sigma) = \sum_{\substack{r_1 \geq r_2 \geq r_3 \geq 0 \\ s_1 \geq s_2 \geq s_3 \geq 0}} z^{r_1} \frac{q^{\sum_{i=1}^3 (r_i^2 - r_i s_i + s_i^2 + \rho_i r_i + \sigma_i s_i) + 2r_3 s_3}}{(q; q)_{r_1 - r_2} (q; q)_{r_2 - r_3} (q; q)_{r_3} (q; q)_{s_1 - s_2} (q; q)_{s_2 - s_3} (q; q)_{s_3} (q; q)_{r_3 + s_3 + 1}}, \quad (2)$$

and $e_3 = (0, 0, 0)$, $e_2 = (0, 0, 1)$, $e_1 = (0, 1, 1)$, $e_0 = (1, 1, 1)$, δ_i the i th unit vector. We suppress z and write $S(\rho|\sigma)$.

Theorem 1. $G_{(3,3,2)}(z, q) = \text{FERM}_3(z, q)$, i.e. Warnaar's Conjecture 2 holds for $k = 3$.

Corollary 2. For all $n \geq 0$,

$$Q_{n,(3,3,2)}(q) = \sum_{\substack{n \geq n_2 \geq n_3 \geq 0 \\ m_1, m_2 \geq 0}} q^{n^2 + n_2^2 + n_3^2 - nm_1 - n_2 m_2 + m_1^2 + m_2^2} \begin{bmatrix} n \\ n_2 \end{bmatrix} \begin{bmatrix} n - n_2 + m_2 \\ m_1 \end{bmatrix} \begin{bmatrix} n_2 \\ n_3 \end{bmatrix} \begin{bmatrix} n_2 + n_3 \\ m_2 \end{bmatrix},$$

a manifestly nonnegative polynomial. In particular Warnaar's positivity conjecture holds for the profile $(3, 3, 2)$ at $d = 8$.

Proof of the corollary. $Q_{n,(3,3,2)} = (q; q)_n [z^n] \text{FERM}_3$; the summand of (??) at $n_1 = n$ has denominator $(q; q)_{n_1} = (q; q)_n$, which cancels (here $n_2 - n_3 + m_3 = n_2 + n_3$ since $m_3 = 2n_3$, and the exponent n_3^2 is absorbed into the displayed quadratic form). $\square$

2 Proof of Theorem ??

The proof chains four proved statements.

Step 1: Warnaar's finite form in the limit

Warnaar (A2 Andrews–Gordon paper, Section 7) defines, for $\sigma = (1, \dots, 1, a)$, $a \in \{-1, 1\}$,

$$F_{n_0, m_0; k}^{(a)}(z, q) = \sum_{\substack{n_1, \dots, n_k \geq 0 \\ m_1, \dots, m_k \geq 0}} \frac{z^{n_1}}{(q; q)_{n_k + m_k}} \prod_{i=1}^k q^{n_i^2 - \sigma_i n_i m_i + m_i^2} \begin{bmatrix} n_{i-1} \\ n_i \end{bmatrix} \begin{bmatrix} m_{i-1} \\ m_i \end{bmatrix},$$

and proves (his Proposition “finite form”, case $a = -1$) the polynomial identity

$$F_{n_0, m_0; k}^{(-1)}(z, q) = \sum_{\substack{n_1, \dots, n_k \geq 0 \\ m_1, \dots, m_{k-1} \geq 0}} \frac{z^{n_1} q^{n_k^2}}{(q; q)_{m_0 - n_1 + m_1}} \begin{bmatrix} n_0 \\ n_1 \end{bmatrix} \prod_{i=1}^{k-1} q^{n_i^2 - n_i m_i + m_i^2} \begin{bmatrix} n_i \\ n_{i+1} \end{bmatrix} \begin{bmatrix} n_i - n_{i+1} + m_{i+1} \\ m_i \end{bmatrix}. \quad (3)$$

Fix $k = 3$ and let $n_0, m_0 \rightarrow \infty$. Each coefficient of $z^n q^N$ on both sides stabilizes ($\begin{bmatrix} n_0 \\ n_1 \end{bmatrix} \rightarrow 1/(q; q)_{n_1}$ and $1/(q; q)_{m_0 - n_1 + m_1} \rightarrow 1/(q; q)_\infty$ coefficientwise). The right side of (??) becomes $\text{FERM}_3(z, q)/(q; q)_\infty$. On the left side the binomial chains telescope: $\begin{bmatrix} n_0 \\ n_1 \end{bmatrix} \begin{bmatrix} n_1 \\ n_2 \end{bmatrix} \begin{bmatrix} n_2 \\ n_3 \end{bmatrix} \rightarrow 1/((q; q)_{n_1 - n_2} (q; q)_{n_2 - n_3} (q; q)_\infty)$ likewise for the m -chain. Writing $(r, s) = (n, m)$ and using $\sigma = (1, 1, -1)$, so that the $i = 3$ exponent is $r_3^2 + r_3 s_3 + s_3^2 = (r_3^2 - r_3 s_3 + s_3^2) + 2r_3 s_3$, we obtain

$$\frac{\text{FERM}_3(z, q)}{(q; q)_\infty} = T(z, q) := \sum_{\substack{r_1 \geq r_2 \geq r_3 \geq 0 \\ s_1 \geq s_2 \geq s_3 \geq 0}} z^{r_1} \frac{q^{\sum_{i=1}^3 (r_i^2 - r_i s_i + s_i^2) + 2r_3 s_3}}{(q; q)_{r_1 - r_2} (q; q)_{r_2 - r_3} (q; q)_{r_3} (q; q)_{s_1 - s_2} (q; q)_{s_2 - s_3} (q; q)_{s_3} (q; q)_{r_3 + s_3}}. \quad (4)$$

Step 2: Pochhammer split

Using $\frac{1}{(q; q)_{r_3 + s_3}} = \frac{1 - q^{r_3 + s_3 + 1}}{(q; q)_{r_3 + s_3 + 1}}$ and $q^{r_3 + s_3 + 1} = q \cdot q^{r_3} \cdot q^{s_3}$ (a shift $(\rho_3, \sigma_3) \mapsto (\rho_3 + 1, \sigma_3 + 1)$ in (??)),

$$T(z, q) = S(e_3 | e_3) - q S(e_2 | e_2). \quad (5)$$

Step 3: the bridging identity is one Kanade–Russell relation

Kanade–Russell (proved; restated as Uncu’s Lemma on recurrences) show that for $m \equiv -1 \pmod{3}$ and $\sigma_{k-1} = 0$,

$$R_3(\rho|\sigma) : S(\rho|\sigma) - S(\rho|\sigma + \delta_3) - q S(\rho + \delta_3|\sigma + \delta_3) + q S(\rho + \delta_3|\sigma + \delta_2 + \delta_3) = 0.$$

At $\rho = \sigma = (0, 0, 0)$ this reads $S(e_3|e_3) - S(e_3|e_2) - q S(e_2|e_2) + q S(e_2|e_1) = 0$, i.e.

$$S(e_3|e_3) - q S(e_2|e_2) = S(e_3|e_2) - q S(e_2|e_1). \quad (6)$$

Step 4: Uncu’s modulo-11 theorem

Uncu (2024, Theorem for $m = 11$) proves, for the canonical profile $(3, 3, 2)$,

$$H_{(3,3,2)}(z, q) = S(e_3|e_2) - q S(e_2|e_1). \quad (7)$$

Conclusion

Combining (??)–(??), $\text{FERM}_3(z, q) = (q; q)_\infty H_{(3,3,2)}(z, q) = G_{(3,3,2)}(z, q)$. □

3 Numerical verification and remarks

Remark 3. *All steps were verified in exact $\mathbb{Z}[q]$ arithmetic (`seed2_R2L3_s11_chain.sage`, precision q^{300} , compared to q^{150}): identity (??) at z -orders $n = 0, \dots, 4$, and $(q; q)_n (q; q)_\infty [z^n] (S(e_3|e_3) - q S(e_2|e_2)) = Q_{n,(3,3,2)}$ against the independently computed Corteel–Welsh ground truth (`seed5_R2L2_qn_d8.json`) for $n = 0, \dots, 4$. Both pass.*

Remark 4 (Extensions). (i) The $a = +1$ case of Warnaar’s finite form should give, by the same chain, the balanced profile $(4, 3, 3)$ at modulus 13 ($d = 10$) via Uncu’s $m = 13$ theorem; the bridging relation is the $m \equiv 1 \pmod{3}$ variant of R_3/R_4 (note: the displayed R_3 for $m \equiv 1$ in Uncu’s paper contains two identical terms of opposite sign, apparently a typo; it should be re-derived from Kanade–Russell Lemma 9.2 before use). (ii) For general k , Warnaar’s finite form reduces the balanced case of Conjecture 2 at every level to the Kanade–Russell/Uncu expression for $H_{(k,k,k-1)}$, which is proved only for $m = 11, 13$. (iii) The Pochhammer merge of Step 2 requires an S -pair differing by $(+\delta_3 | +\delta_3)$ with weight q ; among the $d = 8$ core orbits only $(3, 3, 2)$ acquires this shape after a single R_3 . The remaining core orbits $(6, 1, 1), (5, 1, 2), (4, 1, 3), (4, 3, 1), (5, 2, 1), (4, 2, 2)$ need R_1/R_2 -moves (each costing an extra zq^*S -term) before a merge is possible; this is the proposed template for the frontier.